How well can persistent contrails be predicted over India?

Design and pre-registered evaluation plan for STRATUM, a satellite-corrected contrail forecasting system

Shrugal Vinod · Aditya Harsoda · Ishaan Kupnur · Abeer Shah

Supervisor: Dr. Varsha Pant · Indus International School, Bangalore, India

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1.0	12 October 2026	Methods paper, no results yet	Aerospace & Aviation

READ THIS FIRST This paper fixes our method, data and hypotheses before we look at any results. It reports no experimental findings. Section 7 says in advance what results would support or refute each hypothesis, and we will publish the outcome whichever way it goes.

ABSTRACT

Contrail cirrus is estimated to be the largest single part of aviation's effective radiative forcing, and unlike CO_2 it can often be reduced by small changes in flight altitude. Avoidance depends on predicting ice-supersaturated air at cruise altitude, which weather models do poorly. Validation studies and avoidance trials have so far focused on Europe, the North Atlantic and the United States. For India, one of the fastest-growing aviation markets, a recent global study estimates relatively low contrail forcing per kilometre flown, but that estimate relies on humidity corrections calibrated with measurements from other regions. We describe STRATUM, a system that (i) applies the Schmidt–Appleman criterion and the Contrail Cirrus Prediction Model to open ADS-B trajectories and ERA5 reanalysis over Indian airspace; (ii) detects contrails in Meteosat-9 Indian Ocean Data Coverage imagery and matches them to wind-advected flight segments; and (iii) trains a correction model on those observations, using only scenes in which a contrail could have been seen. For each flagged segment, STRATUM ranks altitude alternatives by estimated warming avoided against extra fuel burned. We pre-register four hypotheses, define four baselines including two published humidity corrections, and specify time-blocked and space-blocked splits to prevent leakage. We also list the main threats to validity: incomplete ADS-B coverage, limited satellite observability under monsoon cloud, coarse reanalysis resolution, and uncertainty in the radiative effect of contrails.

Keywords: contrails · ice supersaturation · ERA5 · Meteosat · ADS-B · India · aviation climate impact

1 Introduction

Aviation warms the climate through its CO_2 and through several shorter-lived effects. The most widely used assessment puts contrail cirrus at 57.4 mW m^{-2} of the 100.9 mW m^{-2} net effective radiative forcing from aviation in 2018, compared with 34.3 mW m^{-2} for CO_2 [6]. Later flight-by-flight modelling revised the 2019 contrail radiative forcing down to 62.1 mW m^{-2}, about 44% below the earlier best estimate, with a sensitivity range of 34.8 to 74.8 mW m^{-2} [23]. The size of the term is uncertain. That it is large is not much in doubt.

What makes contrails interesting from an engineering point of view is that they are concentrated and avoidable. In 2019 about 2% of all flights caused 80% of global contrail energy forcing [23]. A modelling study over Japan found that small altitude changes for a small fraction of flights could cut contrail

energy forcing by up to 59% [21]. In a randomised trial with a US airline, flights whose pilots adjusted altitude on the basis of predictions had 64% fewer contrails than control flights, at a cost of about 2% more fuel on each adjusted flight [18]. Timing matters too: in a UK study, night flights were about a quarter of daily traffic but produced 60–80% of contrail forcing, because a contrail at night traps heat without reflecting any sunlight [19].

The weak link is the forecast. Contrails persist only in air supersaturated with respect to ice, and humidity at cruise altitude is hard to measure and hard to model [3]. Compared with in-situ aircraft measurements, ERA5 has a dry bias of up to −5.5% in relative humidity at 250 hPa, and it flags persistent-contrail conditions in 7.9% of cases where the measurements show 12.1% [24]. Published systems therefore correct ERA5 humidity before use [11, 22, 23], and the humidity correction turns out to be the most influential input in global estimates [23]. Those corrections are fitted to in-situ data from commercial aircraft programmes that sample some regions far more than others; the North Atlantic correction of [22] was later extended to other latitudes with fitted functions of latitude [23].

India is a clear gap. It is now the third-largest domestic aviation market, its fleet more than doubled between 2014 and 2025, and domestic traffic has grown about 9% a year for a decade [10]. The one global study that covers it finds that India and China together account for 19% of flight distance but only 6% of contrail energy forcing, and attributes this to the tropical circulation and lower cruise altitudes [23]. That may well be correct. But the satellite work used to check contrail predictions has concentrated on GOES-16 over the Americas [9] and Meteosat over Europe and the North Atlantic [13]. We have not found a study that tests contrail forecasts over Indian airspace against observed contrails. Until someone does, the low estimate for India is a model result that has not been checked.

This paper describes how we intend to check it. Our contributions are:

1. An open pipeline that produces segment-level contrail predictions for Indian airspace from ADS-B, ERA5 and CoCiP.
2. A hand-labelled set of contrails in Meteosat-9 Indian Ocean Data Coverage imagery, with measured agreement between labellers.
3. A humidity correction trained on observed contrails rather than on in-situ humidity, which restricts training to scenes where a contrail would have been visible.
4. A pre-registered evaluation against four baselines, with time-blocked and space-blocked splits.
5. A decision layer that ranks flights by the lower bound of net warming avoided, so that an alert is only raised when it is likely to help.

Section 2 summarises the physics and related work. Section 3 describes the data, Section 4 the methods, Section 5 the system, and Section 6 the evaluation and hypotheses. Sections 7 and 8 set out what results would mean and what could go wrong.

2 Background

2.1 Contrail formation: the Schmidt–Appleman criterion

Exhaust leaving an engine is hot and moist. As it mixes with cold ambient air, the temperature T and water vapour partial pressure e of the plume move along a straight line in the (T, e) plane [1, 15, 16]. The slope of that mixing line is

$$G = EI_{H_2O} \cdot c_p \cdot p / [\, \varepsilon \cdot Q \cdot (1 - \eta)\,] \tag{1}$$

where $EI_{H_2O} \approx 1.23$ kg kg^{-1} is the water vapour emission index of kerosene, $c_p = 1004$ J kg^{-1} K^{-1} is the specific heat of air, p is ambient pressure, $\varepsilon = 0.622$ is the ratio of the molar masses of water and dry air, $Q = 43.2$ MJ kg^{-1} is the fuel's specific combustion heat, and η is the overall propulsion efficiency of the aircraft [16]. A contrail forms if, somewhere along the mixing line, the plume becomes saturated with respect to liquid water: droplets condense and then freeze. The limiting case is the line that just touches the water saturation curve $e_w(T)$. Its tangent point defines the threshold temperature T_{LM}:

$$d e_w / dT \,|_{T = T_{LM}} = G \tag{2}$$

At 250 hPa (about FL340) with $\eta = 0.3$, equation (1) gives $G \approx 1.64$ Pa K^{-1}. Solving equation (2) numerically with the Murphy and Koop saturation formulas [8] gives $T_{LM} \approx -41.9$ °C. Ambient air colder than this forms a contrail even when completely dry; slightly warmer air forms one only if it is humid enough. Newer engines with higher η have a steeper mixing line and form contrails at slightly warmer temperatures [16].

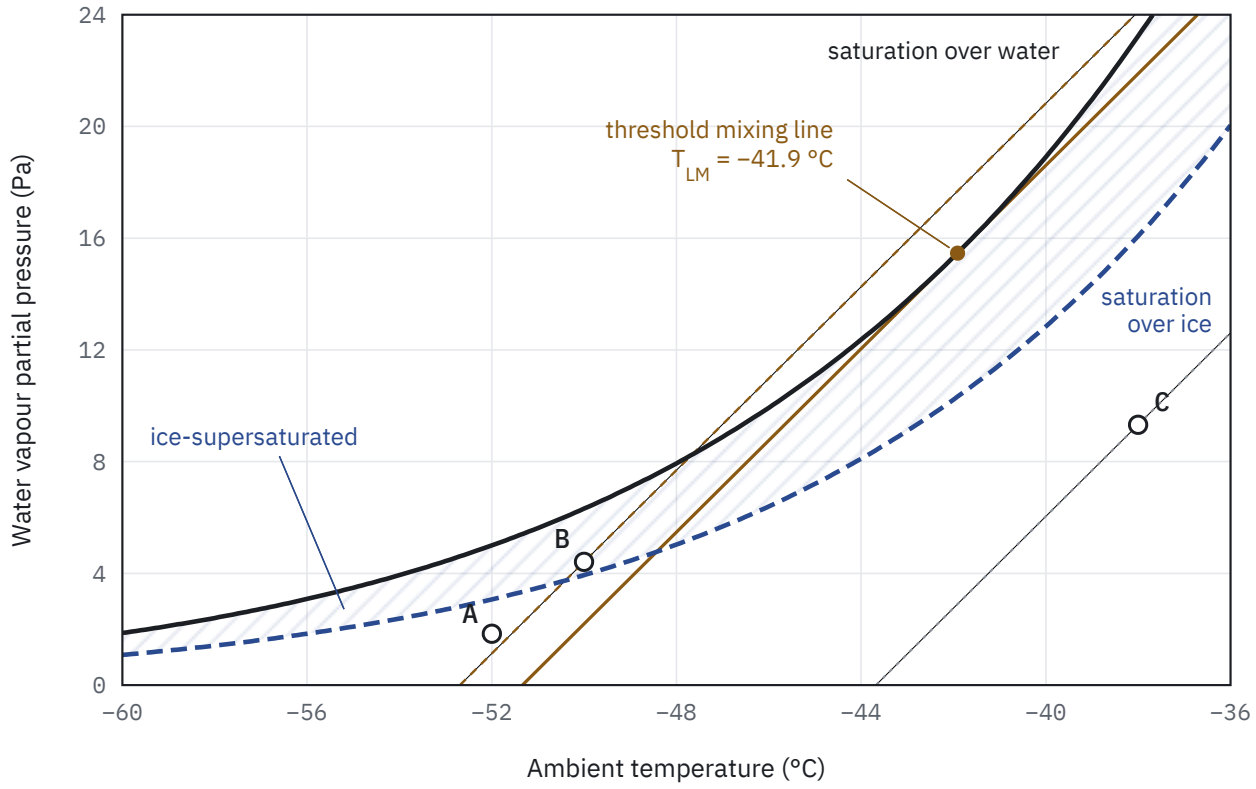


Figure 1. Schmidt–Appleman diagram at 250 hPa with $\eta = 0.3$, computed from equations (1)–(2) and the Murphy and Koop saturation curves [8]. The solid brown line is the threshold mixing line; it touches water saturation at −41.9 °C. Hatched: air supersaturated with respect to ice but not water. Point A (−52 °C, RHi 60%) forms a contrail that evaporates within minutes. Point B (−50 °C, RHi 112%) forms a persistent contrail; its mixing line (dashed) crosses water saturation. Point C (−38 °C, RHi 58%) is too warm and too dry to form one.

2.2 Persistence: ice-supersaturated regions

Once ice crystals form, what happens next depends on the ambient relative humidity with respect to ice (RHi). Below 100% the crystals sublimate and the contrail disappears within minutes. At or above 100% they survive, take up more water, and the contrail can spread into a cirrus sheet that lasts for hours [5]. These ice-supersaturated regions (ISSRs) are patchy and often shallow compared with the vertical spacing between flight levels. That is why a change of one or two flight levels can be enough to avoid

them, and also why a model with coarse vertical resolution struggles to place them. Part of ERA5's difficulty is structural: its cloud scheme limits humidity under cloudy conditions, which suppresses supersaturation [24].

2.3 Climate effect of a contrail

A contrail traps outgoing longwave radiation and reflects incoming sunlight. The net effect depends on the time of day, the surface and clouds beneath it, the contrail's optical depth and how long it lives. CoCiP follows each contrail segment as it is advected, spreads and sediments, and reports its energy forcing (EF, in joules): the radiative forcing integrated over the contrail's area and lifetime [17]. At night only the warming term applies, which is why night flights carry so much of the total [19].

2.4 Related work

Table 1. Studies STRATUM builds on.

STUDY	REGION / SENSOR	WHAT IT DID	HOW WE USE IT
Schumann 2012 [17]	Global	Contrail Cirrus Prediction Model (CoCiP), segment-level contrail evolution and EF	Physics engine, through pycontrails [12]
Teoh et al. 2020 [21]	Japan	Small altitude diversions; up to 59% reduction in contrail EF	Motivates ±2,000 ft alternatives
Teoh et al. 2022, 2024 [22, 23]	N. Atlantic; global	Flight-level contrail EF with latitude-dependent ERA5 humidity correction	Baseline B3; only existing India estimate
Wolf et al. 2025 [24]	IAGOS flight regions	Bivariate quantile mapping of ERA5 temperature and humidity	Optional baseline B4; bias figures
Platt et al. 2024 [11]	ERA5 ensemble	Humidity uncertainty propagated through CoCiP; per-flight skill 84% vs 44% for climatology	Uncertainty method
Ng et al. 2023 [9]	GOES-16 ABI	OpenContrails labelled detection benchmark	Model for our labelling protocol
Santos Gabriel et al. 2026 [13]	Meteosat SEVIRI	Adaptive contrail detection with a confidence score	Detector design for SEVIRI
Driver et al. 2025 [2]	Synthetic imager	Only about 46% of modelled contrail segments observable at 2 km resolution	Observability mask; label bias
Sonabend-W et al. 2024 [18]	United States	Randomised avoidance trial: 64% fewer contrails, about 2% more fuel per adjusted flight	Evidence that avoidance works in operations

3 Study area and data

3.1 Domain and period

The study domain is 6–37°N, 68–98°E, which covers Indian domestic airspace, and we consider cruise segments above FL250 (roughly 376 hPa). We will develop the pipeline on a single corridor, Delhi–Mumbai, one of the busiest domestic routes in the country, before expanding to the full domain. The

development month is January 2026, chosen because the post-monsoon and winter months have clearer skies and therefore more observable contrails. The evaluation period is March 2025 to February 2026, which covers the pre-monsoon, monsoon, post-monsoon and winter seasons once each.

3.2 Datasets

Table 2. Data sources. All are free for research use.

DATASET	VARIABLES	RESOLUTION	KNOWN ISSUES
OpenSky Network [14]	ADS-B position, barometric altitude, velocity, callsign, ICAO24 address	seconds, where covered	Receivers concentrated in Europe and the US; coverage of India is unknown and must be measured
ADS-B Exchange	As above	5 s snapshots; free on the 1st of each month	Only 12 free days a year
ERA5 pressure levels [4]	T, specific humidity, u, v, vertical velocity, cloud fraction, cloud ice water content	0.25°, hourly, 37 levels	Dry bias near cruise altitude; supersaturation limited under cloud [24]
Meteosat-9 IODC SEVIRI	Brightness temperatures at 10.8 and 12.0 μm, plus other channels	about 3 km at nadir, every 15 min, from 45.5°E	Pixels grow at high viewing angles over eastern India; young contrails narrower than a pixel
OpenAP [20]	Aircraft performance, fuel flow	per aircraft type	Less precise than licensed performance models

3.3 Coverage audit and decision rule

Before modelling anything, we will measure how much of the corridor's traffic each flight data source captures. For a sample of days we will count flights with a continuous cruise track (no gap longer than 10 minutes) and compare with the number of scheduled flights on the route. We commit now to the following rule. If OpenSky captures at least half of scheduled corridor flights with continuous cruise tracks, we use OpenSky for all months. If not, we use the 12 free ADS-B Exchange days, accept the smaller sample, and say so in every result.

4 Methods

4.1 Overview

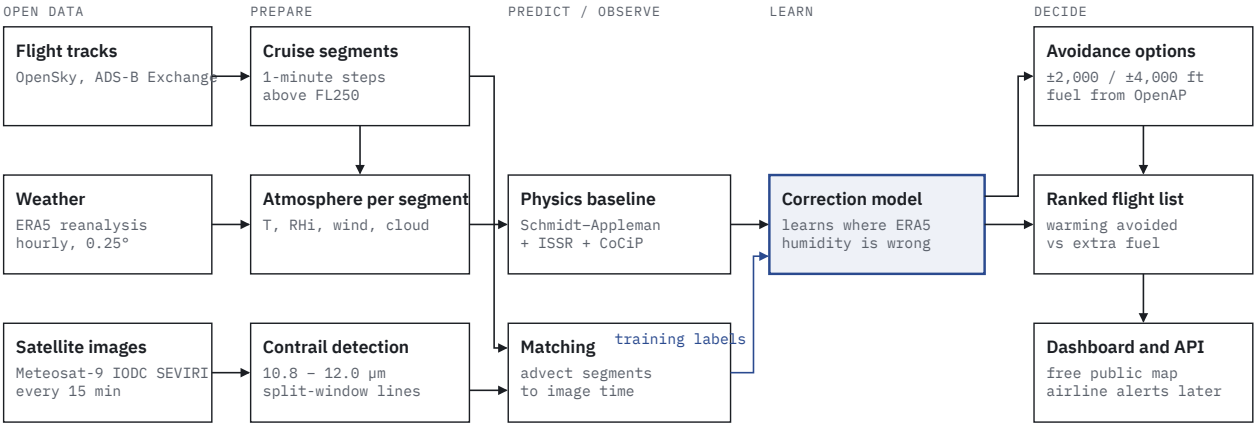


Figure 2. The STRATUM pipeline. Satellite detections, matched to wind-advected flight segments, provide training labels for the correction model (blue). The physics baseline on its own is baseline B1 in Section 6.

4.2 Trajectory processing

We keep airborne state vectors at or above FL250 and remove physically implausible points (vertical rate above 6,000 ft min^{-1} or ground-speed jumps above 100 kt between consecutive reports). Tracks are resampled to one-minute segments. Gaps of up to ten minutes are filled along the great circle at constant altitude, and longer gaps split the flight. Each aircraft's type comes from its ICAO24 address via the OpenSky aircraft database, and its engine, η and fuel flow from OpenAP [20], with aircraft mass set to a fixed fraction of maximum take-off mass and varied in sensitivity tests.

4.3 Atmospheric state and physics baseline

ERA5 fields are interpolated to each segment: linearly in time, linearly in log-pressure in the vertical, and bilinearly in the horizontal. RHi is computed from specific humidity, temperature and pressure using the Murphy and Koop ice saturation formula [8]. For each segment we apply equations (1) and (2) with the aircraft's η, flag persistence where RHi $\geq$ 100%, and run CoCiP through pycontrails [12] to obtain the contrail's lifetime, optical depth and EF.

4.4 Contrail detection in Meteosat-9 imagery

Ice crystals absorb more strongly at 12.0 µm than at 10.8 µm, so a thin contrail shows up as a narrow, line-shaped positive anomaly in the brightness temperature difference BTD = $T_{10.8} - T_{12.0}$ [7]. Because this uses thermal infrared only, it works by day and by night, which matters given how much of the forcing happens at night. We will normalise the BTD image, apply oriented line filters, keep linear objects above a minimum length, and give each detection a confidence score in the spirit of [13], so that precision can be traded against recall.

For validation we will hand-label 200 scenes of 256 × 256 pixels. Scenes are sampled across months, times of day and predicted contrail density, so that the set is not dominated by easy, clear-sky cases. Each scene is labelled by two team members working independently; we report Cohen's κ at object level and settle disagreements by discussion. The first draft of the labelling guide (Appendix B) was written with help from our AI assistant. Before full labelling begins, both labellers will work through a pilot round of 20 scenes, and the guide will be revised wherever the pilot shows it is ambiguous. We also record the satellite viewing zenith angle for each scene, because eastern India is seen at a steep angle from 45.5°E and detection there will be weaker.

4.5 Matching predictions to observations

For a flight segment flown at time t_0, we advect its position with ERA5 winds at its pressure level to each image time t with contrail age $t - t_0$ between 20 and 120 minutes. Younger contrails are usually narrower than a pixel; older ones have spread and drifted enough to be ambiguous. A segment is labelled *observed* ($y = 1$) if a detected linear object lies within two pixels of the advected segment and its orientation is within 20° of the segment's heading at that time.

The negative label needs more care. Satellites miss many contrails even in ideal conditions: one study estimated that only about 46% of modelled contrail segments are observable with a 2 km imager, and fewer than 2% when there is a cirrus layer behind them [2]. SEVIRI's pixels are larger still. We therefore define a scene as *observable* only if ERA5 high cloud fraction at and above the segment's level is below 0.5, the satellite cloud mask shows no opaque cloud within the matching window, and the viewing zenith angle is below 60°. A segment with no matched detection in an observable scene is labelled $y = 0$; in an unobservable scene its label is missing and it is excluded from training and scoring. Even then, $y = 0$ means "not detected", not "no contrail". The model therefore predicts the probability of a *detectable* persistent contrail. Converting that into a probability of any persistent contrail requires a detection efficiency, which we will estimate from CoCiP optical depths and widths using the thresholds in [2], and report as a separate, clearly marked source of uncertainty.

4.6 Correction model

The correction model is trained on segments that satisfy the formation criterion, since a humidity correction can only change persistence. Its features are: ERA5 RHi at the segment and at the levels above and below; the vertical RHi gradient; temperature and its margin below T_{LM}; pressure; cloud ice water content and high cloud fraction; vertical velocity and wind shear; distance to the tropopause (WMO lapse-rate definition from the ERA5 temperature profile); latitude and longitude; and month and local solar time, both encoded cyclically.

The main model is gradient-boosted trees, constrained so that predicted probability can only increase with RHi and decrease as the air gets warmer relative to T_{LM}. Probabilities are calibrated with isotonic regression on a validation fold. As a sanity check we also fit a logistic regression on RHi alone; if the boosted model cannot beat it, the extra features are not earning their place. We will test two ways of feeding the correction back into the physics: (a) using the predicted probability directly for ranking, and (b) learning a scaling factor for RHi in each season and latitude band, then rerunning CoCiP with the scaled humidity to get corrected EF. Option (b) is simpler and easier to explain, so we will prefer it unless (a) is clearly better.

4.7 Avoidance options and the fuel–warming tradeoff

For each segment the model flags, we evaluate alternatives at ±2,000 and ±4,000 ft and rerun CoCiP for each. Extra fuel comes from OpenAP, including an approximate cost for the climb or descent. Fuel converts to CO_2 at 3.16 kg per kg of fuel. Comparing a contrail's warming with a quantity of CO_2 needs a climate metric, and the answer depends on which one is used and over what time horizon. We will therefore report the comparison under both a 20-year and a 100-year horizon and rank flights by the *lower bound* of net warming avoided across the humidity, detection and metric uncertainties. An alert is shown only when that lower bound is positive. This is deliberately conservative: a false alarm burns fuel for nothing and would quickly destroy a dispatcher's trust.

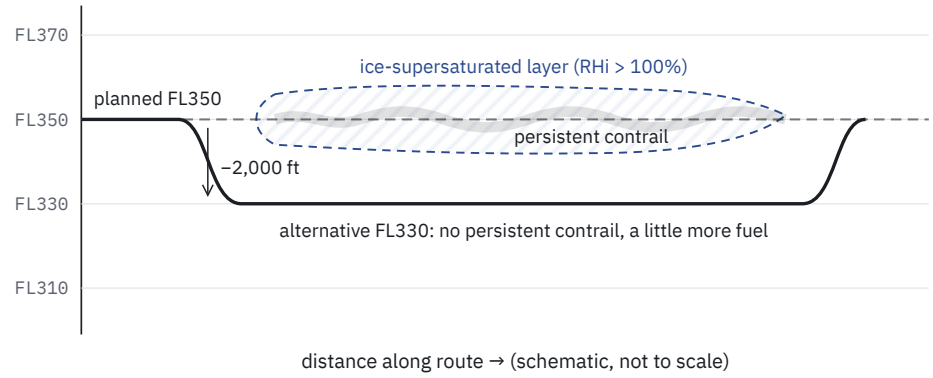


Figure 3. The decision STRATUM evaluates for each flagged segment: stay at the planned level through an ice-supersaturated layer, or change level to avoid it at a small fuel cost.

5 System design

STRATUM's public output is a web dashboard built from archived data, updated daily once the pipeline is stable. It has four views: a map of persistent-contrail probability by flight level; the day's flights drawn over it; a ranked list of flagged segments; and a card for each segment with the fields in Table 3. A methods page explains the model and links to the evaluation results. The front end is a static site, so it can be hosted for free; heavy computation happens offline.

Table 3. Fields shown for each flagged segment. This is the output format only; no values have been computed yet.

FIELD	MEANING
Flight, route, segment time (UTC)	Which flight and where along it the contrail risk is
Planned flight level	Level in the ADS-B track
P(persistent contrail)	Calibrated probability from the correction model
Contrail EF, range	CoCiP energy forcing under corrected humidity, 5th–95th percentile
Best alternative level	The ±2,000 or ±4,000 ft option with the largest lower-bound benefit
Extra fuel (kg) and CO_2 (kg)	Cost of the alternative from OpenAP
Net warming avoided, range	Under 20- and 100-year horizons
Confidence note	Whether the scene type is well covered by validation data

6 Evaluation and pre-registered hypotheses

6.1 Baselines

Table 4. Baselines. Each is scored on exactly the same segments and labels as STRATUM.

ID	BASELINE	DESCRIPTION
B0	Climatology	Monthly frequency of persistent-contrail conditions in each grid cell and flight level, from training months
B1	Raw ERA5	Schmidt–Appleman and RHi ≥ 100% with no humidity correction

ID	BASELINE	DESCRIPTION
B2	Constant scaling	ERA5 RHi divided by 0.95, the simple correction used as a sensitivity case in [23]
B3	Latitude-dependent correction	The correction of [22, 23] as implemented in pycontrails [12]
B4	Quantile mapping (optional)	The bivariate quantile mapping of [24], if its coefficients can be applied to our domain

6.2 Splits and leakage control

Segments from the same flight, and detections from the same satellite image, never appear on both sides of a split. The main evaluation uses four seasonal folds (March–May, June–September, October–November, December–February): we train on three seasons and test on the fourth, rotating through all four. A second check trains on segments west of 80°E and tests on segments east of it, which also tells us how much the steeper viewing angle in the east matters. Thresholds and model settings are tuned only on validation data carved out of the training seasons.

6.3 Metrics

We report hit rate (probability of detection), false alarm ratio, critical success index, AUROC, Brier score and reliability diagrams. The primary metric, fixed now so that we cannot pick a favourable one later, is the **Brier skill score of STRATUM relative to B3 on held-out seasons**. Confidence intervals come from a bootstrap that resamples whole days (1,000 resamples), since segments on the same day share weather and are not independent.

6.4 Hypotheses

Table 5. Pre-registered hypotheses, with the result that would count as support.

ID	HYPOTHESIS	SUPPORTED IF
H1	Raw ERA5 (B1) under-predicts observed persistent contrails over India in observable scenes.	The 95% interval for the ratio of predicted to observed occurrence lies entirely below 1. An interval that includes 1 counts as not supported.
H2	The STRATUM correction improves on the best published correction (B3).	Mean Brier skill score against B3 is at least 0.10 on held-out seasons and its 95% interval excludes 0.
H3	Contrail warming over the corridor is concentrated in a small share of flying.	No more than 10% of cruise flight distance accounts for at least 80% of positive modelled EF.
H4	Night flying carries a disproportionate share of contrail warming over India.	Segments flown between 18:00 and 06:00 local solar time have a larger share of positive EF than of flight distance, with the 95% interval of the difference excluding 0.

We will publish the outcome for every hypothesis, including those that fail. A refuted H1 would be a useful result in itself: it would support the existing low estimate for India rather than overturn it.

7 What the results would mean

Table 6. Possible outcomes and what we would do next.

OUTCOME	INTERPRETATION	NEXT STEP
H1 and H2 supported	ERA5 misses contrails over India, and observations can fix much of it. Existing estimates for India are probably too low.	Recompute EF for the full domain with the correction; publish the revised estimate with its uncertainty.
H1 supported, H2 not	The bias is real but our features cannot correct it.	Publish the bias measurement; look at better humidity information such as in-situ data on Indian routes.
H1 not supported	ERA5 does about as well over India as elsewhere; the low estimate stands.	Shift focus to H3 and H4: finding the few flights and night routes where avoidance still pays.
Too few labels	Fewer than 300 matched observed contrails in observable scenes; H1 and H2 cannot be tested with useful power.	Report the physics-only system with ranges; expand labelling or add a second satellite.

8 Threats to validity and limitations

Table 7. Main risks to the conclusions, and how the design responds.

RISK	EFFECT	MITIGATION
Patchy ADS-B coverage	Missing flights; biased sample of routes	Coverage audit and decision rule (Section 3.3); coverage reported with every result
Limited satellite observability	Labels favour thick, long-lived contrails; monsoon months under-represented	Observability mask; unobservable treated as missing; results reported by season and viewing angle
Wind advection error	True contrails fail to match, or false matches occur	Age window, distance and angle tolerances, with a sensitivity analysis on each
Labeller subjectivity	Noisy ground truth	Written guide, double labelling, Cohen's κ reported
Coarse reanalysis	Thin ISSRs smeared across levels	Neighbouring-level features; results by flight level
Radiative uncertainty	Warming estimates could be off by a large factor	Ranges, two metric horizons, ranking on lower bounds
Operational realism	Air traffic control may not allow a suggested level	Out of scope; stated plainly in all outputs
Overfitting and leakage	Optimistic scores	Blocked splits; primary metric and hypotheses fixed here

Two further limitations are worth stating. First, all of this is computational: we will not have changed any real flight, and nothing in STRATUM should be read as a verified emissions reduction. Second, we are a small student team with free-tier computing, which is why the work starts with one corridor and one year.

9 Ethics, openness and use of AI

Data. ADS-B messages are broadcast openly, but they still identify aircraft. The public dashboard will show commercial airline flights only, and we will not publish tracks of private or state aircraft. Where data licences restrict redistribution of raw tracks, we will release our derived labels, segment-level predictions and the scripts to re-download the raw data.

Openness. Code will be released under an open licence, together with the labelled scenes, the evaluation results and a list of the things that did not work.

Use of AI. The paper was written by the authors. We used an AI assistant throughout the project as a study companion: to find papers, to explain physics we had not met before (for example why the mixing line is straight and how its slope is derived), to check our calculation of G and T_{LM}, to help draft the first labelling guide, and to comment on drafts of this paper. We treated its output as suggestions to verify, not as facts. Every cited paper was read by at least one author, all numbers were rechecked against the original sources, and Figure 1 was computed from our own code.

10 Conclusion

Contrail avoidance is one of the cheapest climate actions available to aviation, and the evidence that it works has been building over Europe and the United States. Over India, current estimates say the problem is smaller per kilometre, but no one has checked those estimates against observed contrails. STRATUM is designed to do that check honestly, with the method and hypotheses fixed before any data are examined, and then to turn whatever it finds into a ranked, conservative list of flights where a change of altitude is likely to help. Whether the answer is that India's contrails are underestimated or that the existing estimate holds up, the result will be the first observational test for one of the world's fastest-growing aviation markets.

– Author contributions

Shrugal Vinod: conceptualisation, methodology, correction model, evaluation design, writing. **Aditya Harsoda:** flight data processing, coverage audit, fuel modelling, writing (review). **Ishaan Kupnur:** satellite processing, contrail detection, labelling protocol, writing (review). **Abeer Shah:** system and dashboard design, market context, figures, writing (review). **Dr. Varsha Pant:** supervision and review.

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– References

[1] Appleman, H. (1953). The formation of exhaust condensation trails by jet aircraft. *Bulletin of the American Meteorological Society*, 34, 14–20.

[2] Driver, O. G. A., Stettler, M. E. J., & Gryspeerdt, E. (2025). Factors limiting contrail detection in satellite imagery. *Atmospheric Measurement Techniques*, 18, 1115–1134.

[3] Gierens, K., Matthes, S., & Rohs, S. (2020). How well can persistent contrails be predicted? *Aerospace*, 7(12), 169.

[4] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146, 1999–2049.

[5] Kärcher, B. (2018). Formation and radiative forcing of contrail cirrus. *Nature Communications*, 9, 1824.

[6] Lee, D. S., et al. (2021). The contribution of global aviation to anthropogenic climate forcing for 2000 to 2018. *Atmospheric Environment*, 244, 117834.

[7] Mannstein, H., Meyer, R., & Wendling, P. (1999). Operational detection of contrails from NOAA-AVHRR data. *International Journal of Remote Sensing*, 20(8), 1641–1660.

[8] Murphy, D. M., & Koop, T. (2005). Review of the vapour pressures of ice and supercooled water for atmospheric applications. *Quarterly Journal of the Royal Meteorological Society*, 131, 1539–1565.

[9] Ng, J. Y.-H., McCloskey, K., Cui, J., et al. (2023). OpenContrails: Benchmarking contrail detection on GOES-16 ABI. arXiv:2304.02122.

[10] News on AIR (2025, 20 December). India emerges as world's third largest domestic aviation market. All India Radio, newsonair.gov.in.

[11] Platt, J., Shapiro, M., Engberg, Z., et al. (2024). The effect of uncertainty in humidity and model parameters on the prediction of contrail energy forcing. *Environmental Research Communications*, 6, 095015.

[12] pycontrails contributors (2023–). pycontrails: Python library for modelling aviation climate impacts. Open-source software, github.com/contrailcirrus/pycontrails.

[13] Santos Gabriel, V., Bugliaro, L., Piontek, D., Ries, S., & Voigt, C. (2026). An adaptive segmentation approach for contrail detection in Meteosat Second Generation satellite imagery. *Atmospheric Measurement Techniques*, 19, 3271–3289.

[14] Schäfer, M., Strohmeier, M., Lenders, V., Martinovic, I., & Wilhelm, M. (2014). Bringing up OpenSky: A large-scale ADS-B sensor network for research. *Proceedings of IPSN '14*, 83–94.

[15] Schmidt, E. (1941). Die Entstehung von Eisnebel aus den Auspuffgasen von Flugmotoren. *Schriften der Deutschen Akademie der Luftfahrtforschung*, 44, 1–15.

[16] Schumann, U. (1996). On conditions for contrail formation from aircraft exhausts. *Meteorologische Zeitschrift*, 5, 4–23.

[17] Schumann, U. (2012). A contrail cirrus prediction model. *Geoscientific Model Development*, 5, 543–580.

[18] Sonabend-W, A., Elkin, C., Dean, T., et al. (2024). Feasibility test of per-flight contrail avoidance in commercial aviation. *Communications Engineering*, 3. doi:10.1038/s44172-024-00329-7

[19] Stuber, N., Forster, P., Rädel, G., & Shine, K. (2006). The importance of the diurnal and annual cycle of air traffic for contrail radiative forcing. *Nature*, 441, 864–867.

[20] Sun, J., Hoekstra, J. M., & Ellerbroek, J. (2020). OpenAP: An open-source aircraft performance model for air transportation studies and simulations. *Aerospace*, 7(8), 104.

[21] Teoh, R., Schumann, U., Majumdar, A., & Stettler, M. E. J. (2020). Mitigating the climate forcing of aircraft contrails by small-scale diversions and technology adoption. *Environmental Science & Technology*, 54(5), 2941–2950.

[22] Teoh, R., et al. (2022). Aviation contrail climate effects in the North Atlantic from 2016 to 2021. *Atmospheric Chemistry and Physics*, 22, 10919–10935.

[23] Teoh, R., Engberg, Z., Schumann, U., Voigt, C., Shapiro, M., Rohs, S., & Stettler, M. E. J. (2024). Global aviation contrail climate effects from 2019 to 2021. *Atmospheric Chemistry and Physics*, 24, 6071–6093.

[24] Wolf, K., Bellouin, N., Boucher, O., Rohs, S., & Li, Y. (2025). Correction of ERA5 temperature and relative humidity biases by bivariate quantile mapping for contrail formation analysis. *Atmospheric Chemistry and Physics*, 25, 157–181.

A Appendix A. Constants used in equations (1) and (2)

SYMBOL	QUANTITY	VALUE	NOTE
EI_{H_2O}	Water vapour emission index, kerosene	1.23 kg kg^{-1}	[16]
c_p	Specific heat of air at constant pressure	1004 J kg^{-1} K^{-1}	
ε	Ratio of molar masses, water to dry air	0.622	
Q	Specific combustion heat, kerosene	43.2 MJ kg^{-1}	[16]
η	Overall propulsion efficiency	0.3 (default)	Per type from OpenAP; 0.25–0.40 in sensitivity tests
p	Ambient pressure (example)	250 hPa	About FL340
G	Mixing-line slope, from (1)	1.64 Pa K^{-1}	Our calculation
T_{LM}	Threshold temperature, from (2)	−41.9 °C	Our calculation, Murphy–Koop curves [8]

B Appendix B. Labelling guide (draft version 0.1)

1. Label a contrail only if it is a line at least 10 pixels long in the BTD image and is clearly narrower than it is long.
2. Label each contrail as one polyline along its centre. If two contrails cross, label both.
3. Do not label linear cloud edges, coastlines or terrain features that appear in the 10.8 µm image as well.
4. Mark each scene as *clear*, *partly obscured* or *obscured*. Obscured scenes are excluded from scoring.
5. When unsure, mark the line as *uncertain* rather than leaving it out. Uncertain lines are scored separately.
6. Label without looking at the flight tracks or the model's predictions, so that labels stay independent of the thing being tested.

Why rules 3 and 6 exist: sharp coastlines and terrain edges can create line-like features in infrared images that are easy to mistake for contrails, and a labeller who can see the predicted flight tracks is more likely to "find" faint lines where they expect them. Both rules will be reviewed after the pilot round.